

MOHAMMED RUPAWALA

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 github.com/MohammedRupawala

Education

Adani University

Ahmedabad, Gujarat, India

B.Tech in Information and Communication Technology

Aug 2022 – Present

- CGPA: **7/10**
- Coursework: Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks, Machine Learning, Deep Learning, Computer Vision

Technical Skills

Languages: JavaScript, TypeScript, Python, C++, SQL, GO, HTML/CSS

Technologies: Node JS, FastAPI, React JS, NextJS, Tailwind

Tools: Vs Code, Docker, Git, GitHub

Experience

E2M Solutions

Jan 2026 – Present

AI Engineering Intern

- Engineered a multi-tenant SaaS AI platform used by 20+ organizations, featuring schema-per-tenant isolation with centralized tenant orchestration, scalable RAG pipelines for PDFs/docs/URLs, and AI-powered chatbot systems that reduced support workload by 80–90%.
- Built end-to-end AI content generation workflows including automated blog creation, contextual inline image generation/editing, reducing manual content production effort by 90%.

Projects

CUTI | *GO, Linux*

Feb 2026

- Built a high-performance TCP server in Go that can handle 10,000+ users at the same time efficiently, using an event-driven architecture similar to Redis.
- Implemented Redis's communication protocol (RESP) from scratch, enabling faster command processing and improving overall request speed by 3–4x.
- Engineered a crash-safe data persistence system utilizing an Append-Only File design, achieving 99.9% data integrity during server outages.

Deki | *Go, OS, CLI*

Jan 2026

- Built a Unix-like shell in Go with support for piping, I/O redirection, and 10+ built-in commands.
- Integrated external command execution (e.g. git, ls) using os/exec for full CLI functionality.
- Implemented Trie-based autocomplete for efficient prefix matching and command suggestions.

MindSync | *React, Tailwind, FastAPI, TypeScript, PostgreSQL, Redis, LangChain*

Apr 2025

- Built an AI-powered journaling platform using RAG and vector embeddings, enabling context aware search and conversational querying across 100+ journal entries with 6–7s retrieval latency.
- Integrated a Hugging Face-based sentiment analysis model to perform multi-class mood classification on journal entries of up to 1,000+ words, extracting actionable emotional insights.
- Optimized the retrieval and inference pipeline to parse, analyze, and respond to user queries within 7 seconds end-to-end, balancing accuracy and latency for real-world usage.

Hackathons

Hackspire: Finalist (Top 5), Adani University

Apr 2025

- Built a smart queue management system using demand forecasting to predict optimal arrival times, reducing user wait times by 10+ minutes and improving service efficiency.

Odoo Hackathon: Final Round, GIFT City, Gujarat

Jul 2025

- Developed a Q&A platform with question tagging, threaded answers, and voting, enabling efficient knowledge sharing within organizations.

Competitive Programming

LeetCode(THFC): 1650 — Codeforces(Special_One): 1200